



Raisa Fairouz Meem

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Education & Training

MSc in Computer Science | American International University Bangladesh | 01/01/2022 - 19/03/2023 | Dhaka, Bangladesh

Final grade: 3.77/4.0 | Level in EQF 7 | Thesis: Medical Image Classification using different Transfer Learning Models

BSc in Electronics and Communication Engineering | Khulna University of Engineering and Technology | 01/01/2013 - 12/09/2017 | Khulna, Bangladesh

Final grade: 3.01/4.0 | Level in EQF 6 | Thesis: Experimental Setup on Educational Robotics Invention Kit (ERIK)

Work experience

Research engineer | Debug Vision | 01/06/2023 - 31/08/2025 | Dhaka, Bangladesh

Research and Annotate image datasets to train YOLO models for Real-time Object Detection

Research and build appropriate datasets for model training

Head of Internal Affairs | AUS-BAN International Study Edu-Care | 01/09/2020 - 30/09/2021 | Dhaka, Bangladesh

Build and sustain business relationships with institutes around the world.

Participate in business planning.

Projects

Medical Image Classification using different Transfer Learning Models | 01/05/2022 - 20/02/2023

Comparative study of 4 transfer learning models- EfficientNetB7, InceptionResNetV2, NasNetLarge, and ResNet50 for Osteosarcoma Tumour Detection. InceptionResNetV2 achieved the highest accuracy (93.29%), and exhibited the highest precision (0.8658) and recall (0.8658) values among the four models.

Analyse the performance of InceptionResNetV2 for Bone Marrow Cell Detection. The proposed model achieved 96.19% accuracy.

Real-time Object and Human Behaviour Detection using YOLOv8 | 01/06/2023 - 31/08/2025

Train YOLOv8 models to detect objects and suspicious behaviour in real-time in CCTV footage.

The accuracy varied from 91.6% to 98.9% for certain objects, whereas it dropped to 57.8% to 61.3% for human behaviour due to the scarcity of usable datasets.

Sentiment Analysis of Student Review of Online Course using Transformer Attention Mechanism | 01/10/2022 - 31/12/2022

Apply the BERT-base-Uncased model for Language Modelling, incorporating the attention mechanism for sentiment analysis of student reviews of online courses available on the Massive Open Online Course (MOOC) platform.

The class-imbalanced dataset dropped the accuracy to 76% with a high recall value using the sequence classifier.

Domestic Emotional Abuse in Childhood through Adult Retrospective Accounts | 01/05/2024 - 30/09/2024

50 participants took a survey based on the guidelines of healthline.com on domestic mental abuse. The data was utilised using a relational database. Analysis of the data revealed that 13.89% of respondents reported experiencing extremely high levels of domestic emotional abuse.

Summarise the contents found through hyperlinks in WhatsApp chat and return the results in a PDF file.

Publications

Osteosarcoma tumor detection using transfer learning models

Meem, Raisa Fairouz and Hasan, Khandaker Tabin (2023). Clinical Imaging and Case Reports (ISSN 2770-9205). <https://doi.org/10.48550/arXiv.2305.09660>

Bone marrow cytomorphology cell detection using InceptionResNetV2

Meem, Raisa Fairouz and Hasan, Khandaker Tabin (2023). Journal of Carcinogenesis & Mutagenesis. <https://doi.org/10.48550/arXiv.2305.05430>

Domestic Emotional Abuse in Childhood: Study Based on Adult Retrospective Accounts

Meem, Raisa Fairouz, Mim Zakia Tabassum and Hasan, Khandaker Tabin (2025). International Journal of Advanced Multidisciplinary Research and Studies. <https://doi.org/10.62225/2583049X.2025.5.5.5045>

Honours and Awards

Vice Chancellors Award for Best Thesis for Master of Science in Computer Science | American International University Bangladesh | 19/03/2023

Skills

Python (computer programming) | C, C++ C# | database management systems | HTTP, PHP, .NET | computer vision | Stack: TensorFlow, Pandas, Numpy, Jupiter, Plotly, Matplotlib, Python, SQL, Matlab, Maple| Exel, Word, Canva | Tools: Jupyter, Weka, Anaconda, Google Colab, Visual Studio, Android Studio, Adobe Photoshop CC| Git/Github, Docker, Gitlab | Jupyter Notebook, Google Colab, Kaggle Notebook, VS Code and PyCharm | language teaching methods | artificial neural networks | clinical imaging techniques | statistics | perform data analysis | Data Science | Data Collection, Data Processing, Data Analysis, Data Visualisation

Language Skills

Mother tongue(s): Bengali

	Understanding		Speaking		Writing
	Listening	Reading	Spoken production	Spoken interaction	
English	C2	C2	C1	C1	C1

Certifications

Code in Place | Stanford University | online | 03/07/2023

Networks and Memberships

Women in AI (WAI) | 17/09/2024 - Current | Global

Volunteering

Association of Voluntary Actions for Society (AVAS) | 01/10/2022 - Current | Bangladesh

Friendship (NGO) | 01/01/2025 - Current | Bangladesh

Dream-Voluntary Blood Donation Society of KUET | Khulna | 01/01/2013 - Current | Khulna